1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that EMT programs, TP53 mutations, and loss of E-cadherin converge to drive invasive progression, yet imaging or biomarker signals (e.g., radiologic indolence) sometimes fail to identify tumors that later progress. The key puzzle is: under what microenvironmental/imaging conditions does TP53–CDH1–EMT biology translate into early progression? 2) Tier 1 paths used: I built on Core Path 1 (EMT → invasive phenotype) and Core Path 3 (TP53 mutation ↔ E-cadherin/CDH1 prognostic axis). These provide a mechanistic backbone linking TP53 dysfunction to loss of epithelial adhesion (CDH1) and EMT activation as drivers of invasion. I selected these because they most directly explain how intrinsic tumor programs enable dissemination. 3) Tier 2 evidence supporting the claim: the radiologic/invasiveness thread (Core Path 2 and Supporting Path 3) suggests that imaging surrogates of microenvironmental composition can stratify invasion risk; PSA/PSA-density analogues (Supporting Paths 1–2) exemplify how sampling-level biomarkers can miss biologically high-risk lesions, reinforcing the need to combine molecular status with imaging-derived stromal context. 4) Why PFI as the endpoint: the mechanistic question focuses on conversion from localized/indolent to progressive/invasive behavior — progression-free interval is the most direct clinical readout of that transition and is available in TCGA-style datasets. 5) Deliberate exclusions: I excluded ground-glass nodules (GGNs) from the core hypothesis because they were central to the Ontologist's lung-focused radiology narrative but are not among the imaging features available for this TCGA-style breast cancer/imaging plan; similarly I relegated PSA/PSA-density and CIS to secondary_modifiers because they are supporting clinical-biomarker concepts rather than the primary mechanistic drivers. I also avoided adding additional genes beyond the two Tier-1 anchors to keep the hypothesis sharply focused?

Working Hypothesis

Do tumors with TP53 dysfunction and loss of E-cadherin (CDH1) drive EMT-dependent invasion preferentially when the stromal-to-tumor ratio (TSR) is low, thereby exposing a tumor-dominant microenvironment that amplifies adhesion loss and promotes dissemination? If this is the case, TP53-mutant/CDH1-low tumors with low TSR will exhibit shorter progression-free interval (PFI) than tumors that retain CDH1 or have high TSR, defining a high-risk subset that may be missed by imaging alone.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, CDH1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53–CDH1–EMT biology translate into early progression? 2) Tier 1 paths used: I built on Core Path 1 (EMT → invasive phenotype) and Core Path 3 (TP53 mutation ↔ E-cadherin/CDH1 prognostic axis).

CDH1: CDH1–EMT biology translate into early progression? 2) Tier 1 paths used: I built on Core Path 1 (EMT → invasive phenotype) and Core Path 3 (TP53 mutation ↔ E-cadherin/CDH1 prognostic axis).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	KLK3 (PSA / PSA density surrogate)	Supporting modifier from knowledge graph
Secondary	EMT_program	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis using datasets with matched genomic (TP53 status), transcriptomic or protein (CDH

Prediction 2: data, and histopathology-derived TSR, plus longitudinal PFI.

Prediction 3: Digital pathology to quantify TSR and CDH1 protein levels; targeted sequencing or exome data for TP53 mutations.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.
- Direct evidence in the specific cancer type may be limited; the hypothesis extends findings from related contexts.

Alternative Directions

- KLK3 (PSA / PSA density surrogate): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EMT_program: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.